

Case Study 5: Stack Overflow Tag Prediction

VI: Business/Real world Problem

Kaggle Competition.

V2: Business objectives and Constraints

- 1) High precision and Recall for Tag prediction
- 2) Missing or providing incorrect tag could impact Recall.
- 3) No strict latency constraints.

V3: Mapping to an ML problem: Data Overview

Train.zip 2.19 GB ✓ all labels here.

Test, Zip

GM rows. (Id, Title, body, Tags)

All are lowercase with no \t or \n

Body has lot of code.
Tags are space separated.

V4: Mapping to an ML problem: ML problem formulation

$$D = \{x_i, y_i\}$$

↳ set of tags

multi label not multi class

● V5: Mapping to an ML problem: Performance Metrics

In Binary classification :-

Precision

Recall

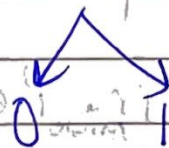
F1

log loss

AUC

● In ^{label}multiclass classification :-

High Precision & Recall use F1 Score (Geometric mean)



For ^{label}multiclass classification :-

We need to modify F1 Score for it.

On Kaggle page:-

1) Mean F1 Score (Micro Average F1 Score)

micro
average
F1 Score

macro
average
F1 Score

✓
for this
case
study.

1) $C = \text{set of tags}$

$\rightarrow t_1, t_2, t_3, \dots, t_n$

~~The Simple F1 score is:-~~

$$F_1 = \frac{2 \cdot \text{Pr}_{\text{micro}} \cdot \text{Re}_{\text{micro}}}{\text{Pr}_{\text{micro}} + \text{Re}_{\text{micro}}}$$

$$\text{Precision}_{\text{micro}} = \frac{\sum_{k \in C} TP_k}{\sum_{k \in C} TP_k + FP_k}$$

$$\text{Recall}_{\text{micro}} = \frac{\sum_{k \in C} TP_k}{\sum_{k \in C} TP_k + FN_k} \quad \left. \vphantom{\frac{\sum_{k \in C} TP_k}{\sum_{k \in C} TP_k + FN_k}} \right\} \begin{array}{l} \text{weighted} \\ \text{F1} \\ \text{Score} \end{array}$$

Micro averaged F1 Score = Takes tag / label freq. of occurrence into consideration.

$t_1 \rightarrow 90\% \uparrow Pr \uparrow Re$

$t_2 \rightarrow 1\% \downarrow Pr \downarrow Re$

So over all F1 Score could be high.

$$\begin{aligned} F1_{\text{macro}} &= \frac{1}{|C|} \sum_{k \in C} F1_{\text{score}_k} \\ &= \frac{1}{|C|} \sum_{k \in C} \frac{2P_k R_k}{P_k + R_k} \quad \left. \vphantom{\frac{1}{|C|} \sum_{k \in C} \frac{2P_k R_k}{P_k + R_k}} \right\} \begin{array}{l} \text{Simple} \\ \text{mean} \\ \text{or} \\ \text{F1 Score} \end{array} \end{aligned}$$

$|C|$ = Size of set C.

It doesn't take frequency of occurrence of tag into consideration.

v6: Hamming Loss

$$\text{Hamming Loss}(x_i, y_i) \\ = \frac{1}{|D|} \sum_{L=1}^{|L|} \frac{\text{xor}(\hat{y}_i, y_i)}{|L|}$$

where $|D|$ = No. of Samples

$|L|$ = " " " " - labels

y_i = ground truth.

$\hat{y}_i$ = prediction.

So, explanation:-

$$L = \{t_1, t_2, t_3, t_4\}$$

Total labels $|L| = 4$

$$Q_1 = t_1, t_2, t_3$$

Create a binary vector for each y_i & $\hat{y}_i$ and summate over all data points.

XOR

0	0	0
0	1	1
1	0	1
1	1	0

$y_i \rightarrow$

1	1	1	0
---	---	---	---

$\hat{y}_i \rightarrow$

1	1	0	0
---	---	---	---

$$\text{XOR of } y_i \text{ \& } \hat{y}_i = 0 + 0 + 1 + 0.$$

(acc. to func).

V7: EDA: Data loading

V8: EDA: Analysis of Tags

V9: EDA: Data Preprocessing

4MM questions, title & body
features

- 1) Sample 1M data points
- 2) Separate out code-snippets from body.
- 3) Remove special characters from ques. title & description (not code)
- 4) Remove stopwords except 'c'

- 5) Remove HTML tags.
- 6) Convert all the characters into small letters
- 7) Use snowball stemmers to stem words.

V10: Data Modelling: Multi label Classification

Binary representation of your labels in a Multiclass setting

t_1, t_2, t_3, t_4

$d_1 \rightarrow t_1, t_2$

1	1	0	0
---	---	---	---

 y_i

$d_2 \rightarrow t_2, t_3$

0	1	1	0
---	---	---	---

$d_3 \rightarrow t_1, t_4$

1	0	0	1
---	---	---	---

of labels = 4.

4 binary classification tasks.

 ~~$D_1 y_1$~~

	y_1
d_1	1
d_2	0
d_3	1

 $D_2 y_2$

	y_2
d_1	1
d_2	1
d_3	0

 $D_3 y_3$

	y_3
d_1	0
d_2	1
d_3	0

 $D_4 y_4$

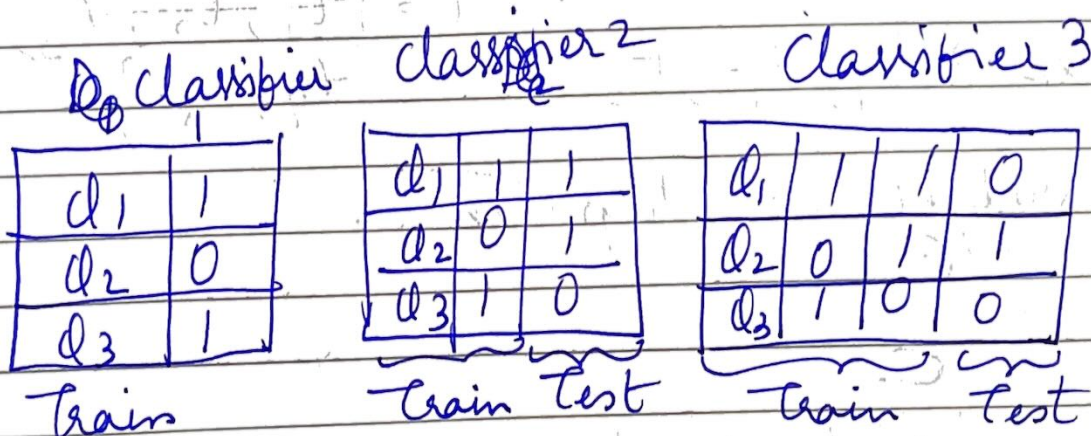
	y_4
d_1	0
d_2	0
d_3	1

One vs. Rest.

of labels is large say 42k
and we have to create 42k
classification tasks for each tags

So, there is a library called
skmultibarn library (Only uses Dense
works when data is small matrices)

Classifier chains



If one label is powerful in
predicting another label.

Label powerset.

X	y_1	y_2	y_3	y_4
x_1	0	1	1	0
x_2	1	0	0	0
x_3	0	1	0	0
x_4	0	1	1	0
x_5	1	1	1	1
x_6	0	1	0	0

→

X	y_1
x_1	1
x_2	2
x_3	3
x_4	1
x_5	4
x_6	3

Modify existing algorithm to multi label.

VII: Data Preparation

Applying Countreducer on tags.

Partial Coverage:

$$d_1 \rightarrow t_1, t_2, t_3$$

$$d_2 \rightarrow t_4, t_5, t_3$$

$$C = \{t_1, t_2, t_3, t_4, t_5\}$$

Subset of tags/labels.

~~$$C' \subseteq C$$~~

$$C' \subset C$$

$$C' = \{t_1, t_2\}$$



$$d_1 \rightarrow t_1, t_2$$

$$d_2 \rightarrow \phi$$

To find smallest subset,

$$t_1 = \text{freq}_1$$

$$t_2 = \text{freq}_2$$

$$t_3 = \text{freq}_3$$

Most frequent 500 tags
90% of the questions could be covered

5500 — 99% of Questions
tags

V12: Train Test Split

● V13: Featurization

TFIDF — Sparse Matrix

V14: Logistic Regression: One vs Rest

SKlearn → OneVsRestClassifier.
Used for multilabel classifier.

● V15: Sampling data and tags + Weighted models

Title is repeated 3 times to give it 3 times more weightage.

V16: Logistic Regression revisited.